Green Drop Certification 2011
REQUIREMENTS

green drop
CERTIFICATION
waste water service
REGULATION



Striving Towards Excellence in South African Wastewater Management

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Legislation

- Constitution
- **Water Services Act**
  - *+secondary legislation*
- **National Water Act**
  - *+secondary legislation*
- Municipal Systems Act

- *The Water Services Regulation Framework defines the regulatory processes.*

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To Note:

1. Requirements may include both legislated **norms and standards** as well as international **best practice**;
2. **Performance** against Requirements be **published** to ensure enhanced levels of accountability at both departmental and municipal level;
    1. *The public have access to credible information and do not have to fall prey to sensationalist reporting;*
3. Performance assessments are conducted on a consultative audit basis which is to advise on required improvements.

Essentials

1. Prepare for the assessment.
2. Provide proof of compliance with set requirements.
3. Ensure adequate Municipal/WSP representation during assessments.

No	Green Drop Criteria 2011	Prim. Weight
1	Process Control, Maintenance and Management Skill	10
2	Wastewater Quality Monitoring Programme	10
3	Wastewater Sample Analysis (credibility)	5 (10)
4	Submission of Wastewater Quality Results	5 (10)
5	Wastewater Quality Compliance	30
6	Wastewater Quality Failures Response Management	10 (20)
7	Storm water and Water Demand Management	0
8	Bylaws	5 (0)
9	Waste Water Treatment Facility Capacity	10
10	Publication of Waste Water Management Performance	0
11	Waste Water Asset Management	15 (0)

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Requirement 1

Operations (Process Control), Maintenance and Management Skill

10%

- To ensure compliance with Regulation 2834;
- Place renewed focus on Operational and Maintenance skills requirements
- To ensure that adequately skilled staff is employed to take charge of process controlling and operations.

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Requirement 1

Operations (Process Control), Maintenance and Management Skill

1.) A copy (certified) of **Registration Certificate of Works** displaying Classification ito (Regulation 2834) ***(20%)***	▪ Copy of the **WWTW registration certificate** must be uploaded on the GDS
2.) Copies (certified) of **Registration Certificates of Process Controllers and Supervisors** (Regulation 2834) ***(40%)***	▪ Copies of the **classification certificates** of **all** process controllers/*operators* and supervisors/superintendents must be uploaded on the GDS ▪ WSI must indicate **shift patterns** ▪ Proof of qualifications and experience of shift workers performing process controlling tasks ▪ Must comply with Reg 2834 ▪ WSI must indicate process controllers *and/or* supervisors that are 'shared' across different plants/sites
3.) Proof of **Maintenance Team** used for general maintenance work at the plant (both mechanical and electrical) ***(10%)***	▪ **Information** on in-house staff or external contractor/s ▪ **Contract** or **Logbook** with maintenance entries will serve as proof ▪ Provide additional proof of competency of team (e.g. Qualification & Experience & Trade-test)

Requirement 1 …*continue*

4.) Proof of a 'site-specific' **Operation & Maintenance Manual** ***(30%)***	▪Copy of front page and index to be provided to DWA, but sufficient content must be proven ▪O&M manual to contain: structural, mechanical, electrical detail of plant, design specifications of plant, reference to drawings, operational schedules, maintenance schedules, process detail and control, instrumentation specification/type, fault finding, monitoring, pump curves, supportive appendices
BONUS: ***(40%)***	Proof of Process Controllers being subjected to relevant training the past 12 months

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Requirement 2

10%

Wastewater Monitoring Programme

- To ensure that adequate monitoring is taking place as a 1st step towards effective wastewater management.
- To promote management & planning based upon measurement/monitoring.

Requirement 2

Wastewater Monitoring Programme

1.) Details of sampling sites; determinands and frequencies of **Operational Monitoring** ***(40%)***	▪ Proof of Operational Monitoring sites, determinands and frequency ▪ Samples must include: inflow, outflow, process flows, industrial effluent, sludge; ▪ Determinands: as per license/permit; Frequency : as per license/permit or as per best practice
2.) Details of sampling sites; determinands and frequencies of **Compliance Monitoring** ***(60%)***	▪ Proof of Compliance Monitoring sites, determinands and frequency ▪ As per specification in license/GA/permit/registration (including Catchment Monitoring, i.e. up/downstream samples) ▪ *Note: for zero-effluent treatment systems - still need to monitor for impact on catchment / environment (for both lined and unlined systems)* ▪ NB: Applicable to oxidation ponds as well

Requirement 3

5%

Wastewater Sample Analysis (credibility)

- To ensure enhanced credibility of analyses.
- To place renewed focus on the role of credibility laboratories.
- To improve the integrity of results (that forms the foundation of so many decisions made in the wastewater sector.
- To serve as basis for the procurement of credible laboratory services.

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Requirement 3

Wastewater Sample Analysis (credibility)

1.) Provide proof and the name of the **Laboratory** used ***(10%)***	▪ Name lab for operational analysis (in-house or on-site) and lab for compliance analysis/checks (in-house or external)
2.) Certificate of **Accreditation** for applicable methods, *Or* Z-scores results following participation a recognised **Proficiency Testing Scheme** ($-2 \geq$ z-score ≥ 2 are unacceptable) *Or* Proof of Intra- and Inter-laboratory proficiency (quality assurance as prescribed in Standard Methods) ***(60%)***	▪ Check if Laboratory is accredited to perform the specific methods, check acceptability of Z-scores for the water quality determinands
3.) Explain how **monitoring results** are used to **amend/improve process controlling** ***(30%)***	▪ Practical example [The assessor will select at random analytical parameter/s from the presented analytical results to present an audit question]
BONUS: Monitoring at an **acceptable frequency** and for the required **determinands** ***(+25%)***	▪ Proof to be provided that WSI maintains a 100% monitoring trend at an acceptable minimum frequency against a full set of required process determinands ▪ *Best practice indicators: low-end techn/small size = 1x-2x/month, medium size = 1x-2x/week, high techn/macro size = 1x/day or hourly*

5%

Requirement 4
Submission of Wastewater Quality Results

- Ensuring that the necessary information/data is submitted to the Department to allow for regulatory performance monitoring to be conducted.
- Essential as per the Section 62 of the Water Services Act (Act 108 of 1997)

Requirement 4
Submission of Wastewater Quality Results

Proof of **data submission** to DWA (12 months) ***(100%)***	▪ 12 months of data submitted to DWA **on the GDS** ▪ WSA must ensure that 12 months' sets of results are submitted and recorded on the GDS prior to the assessment. Note: **All compliance results' data required** ▪ *Green Drop System (GDS) will be operational by 1 Sept 2010 (www.dwa.gov.za/greendrop).* ▪ *Proof is required should hardcopy submissions have been made to DWA Regional Offices.*

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Requirement 5

Effluent Quality Compliance

30%

- To give emphasis to the need for improved Effluent Quality Compliance.
- To ensure that a wastewater treatment facility complies with the limits set in the License/Authorization.
- The most important requirements and weighted accordingly.

Requirement 5

Effluent Quality Compliance

1.) Copy of **effluent quality** limits or standards used to calculate compliance (e.g. effluent limits or standards as per license, GA or permit) ***(20%)***	▪ Copy of applicable Authorisation, containing the specified effluent quality limits or standards for discharge to a water body or for irrigation or for other applications
2.) Effluent Quality CATEGORIES: 90% microbiological compliance; 90% chemical compliance; 90% organoleptic physical compliance ***(80%)***	▪ 90% compliance against all three effluent quality categories (if not authorised – 8x General Limits apply ▪ *Note: 90th percentile compliance considered in case of large datasets to be assessed as performance measure*
BONUS ***(+35%)***	A practical and acceptable Wastewater Management Rectification or Risk Abatement Plan (W_2RAP) is in place to address inefficiencies/inadequacies that result in non-compliance
PENALTY ***(-20%)***	No proof of valid authorisation or sufficient proof of application submitted to DWA

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10%

Requirement 6
Wastewater Quality Failure-Response Management

- To get protocols in place for any incident which may directly or indirectly lead to the compromising of effective wastewater treatment and management.

Requirement 6
Wastewater Quality Failure-Response Management

1.) Proof of a documented **Wastewater Incident Management Protocol** ***(45%)***	▪ Protocol to specify alert levels, response times, required actions, roles & responsibilities and communication measures/vehicles ▪ NB. Include Pump station failure
2.) Provide **evidence of implementation** of Protocol ***(55%)***	▪ Wastewater Quality Failure Incident and Sewer Spillage Incident register

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Requirement 7

Storm Water and Water Demand Management

0%

- To ensure that artificial loading on WWTWs are minimised.
- To minimise the influence of faecal contamination in storm-water and run-off.

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Requirement 7

Storm water and Water Demand Management

Proof of a **Storm-water management plan** detailing how storm-water will be prevented from entering sewer systems and how sewer spillages or sewerage from entering storm-water. Evidence of implementation required	▪ Copy of front page and contents pages + implementation proof
Water Demand Management Plan including a practical strategy to address artificial water demand due to leakages causing higher hydraulic loading of wastewater collection and treatment infrastructure. (Might include the need for a wastewater balance)	▪ Copy of Strategy or Implementation Plan + implementation proof

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Requirement 8
Water Services Bylaws

5%

- To promote local regulation with the purpose of preventing industrial, commercial and domestic practices from having a detrimental impact on the sewer collection and treatment operations.

Requirement 8
Water Services Bylaws

1.) Proof of the Bylaws providing for the regulation of industrial (trade) effluent (volumes & quality) discharged into municipal system, package plants, decentralized systems, vacuum tank discharges and spillages into the environment. ***(40%)***	▪ Copy of front page, index and section referring to industrial/trade effluent
2.) Evidence of Bylaws enforcement by Local Authority ***(60%)***	▪ Proof of application of Bylaw clause in practice, supported by written notice/s to offender

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Requirement 9

Wastewater Treatment Facility Capacity

- To ensure that the treatment and collection capacity risks be mitigated.
- An essential component of Integrated Asset Management

10%

Requirement 9

Wastewater Treatment Facility Capacity

1.) Documented design capacity (**hydraulic and organic**) of the wastewater treatment facility and Documented daily **receiving flows** over the 12months of assessed period (ideally ≤ than design capacity) ***(30%)***	▪ Design capacity as Average Dry Weather Flow (ADWF) and COD load to the plant ▪ Evidence of daily flows and subsequent calculated averages. Measurement method to be explained ▪ Assessor may request proof of calibration certificates of inflow meters to verify accuracy of data
2.) Medium to long term planning to ensure sufficient **capacity for treatment system** and to ensure effluent quality compliance ***(40%)***	▪ Detailed Work-plan which stipulates type of work, associated budget and projected timeframe, as well as the planned output of this work
3.) Medium to long term planning to ensure sufficient **capacity for collecting system** ***(30%)***	▪ Detailed Work-plan which stipulates type of work, associated budget and projected timeframe, as well as the planned output of this work

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Requirement 10

Publication of Wastewater Management Performance

0%

- Enhancing Accountability through enhance transparency.

Requirement 10

Publication of Wastewater Management Performance

Annual Publication of wastewater management performance against the requirements of the site-specific License conditions or General Authorisations ***(40%)***	▪ Name and date of publication, copy of information pertaining to audit question. Note: Level of detail must include compliance detail
Publication in various communication mechanisms to reach wider audience, in particular information to the public ***(60%)***	▪ Evidence/Copy of publication in each media form. Electronic (web) good but not entirely sufficient. Web-based reporting (only) will equate to 40% of this sub-criteria score

Requirement 11

Wastewater Asset Management

15%

- To re-enforce the crucial importance of effective Asset Management

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Requirement 11

Wastewater Asset Management

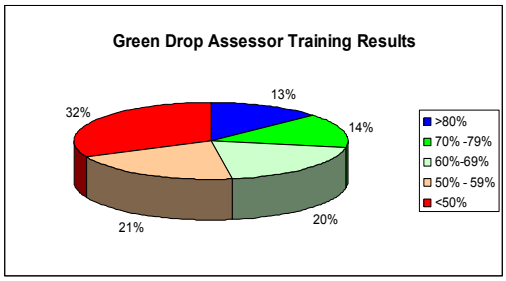
1.) Annual Audit report addressing: collection and treatment infrastructure **and** process control. ***(25%)***	▪ Proof of Technical Audit/ Assessment/ Inspection Report and evidence/plan of implementation of findings during year following Audit Report
2.) Updated sanitation / wastewater Infrastructure Asset **Register** ***(35%)***	▪ Proof of Asset Register, evidence to be submitted. Asset register to include movable equipment and immovable infrastructure / assets (Cover page plus Index)
3.) Operation and maintenance **budget** and comparative **expenditure** detail for: 1 - wastewater treatment (in cents/m^3), and 2 - collection system (R/m^3) ***(20%)***	▪ 50% score to proof of budget, 50% score to proof of expenditure against budget. ▪ Guide 1: low end technology = R0.50/Kl, medium to high end techno = R0.80-1.20/ Kl ▪ Guide 2: R 55 000/(Ml/day plant capacity)
4.) Maintenance records of pump-stations ***(20%)***	▪ Proof of maintenance work done on mechanical, electrical, civil per pump station

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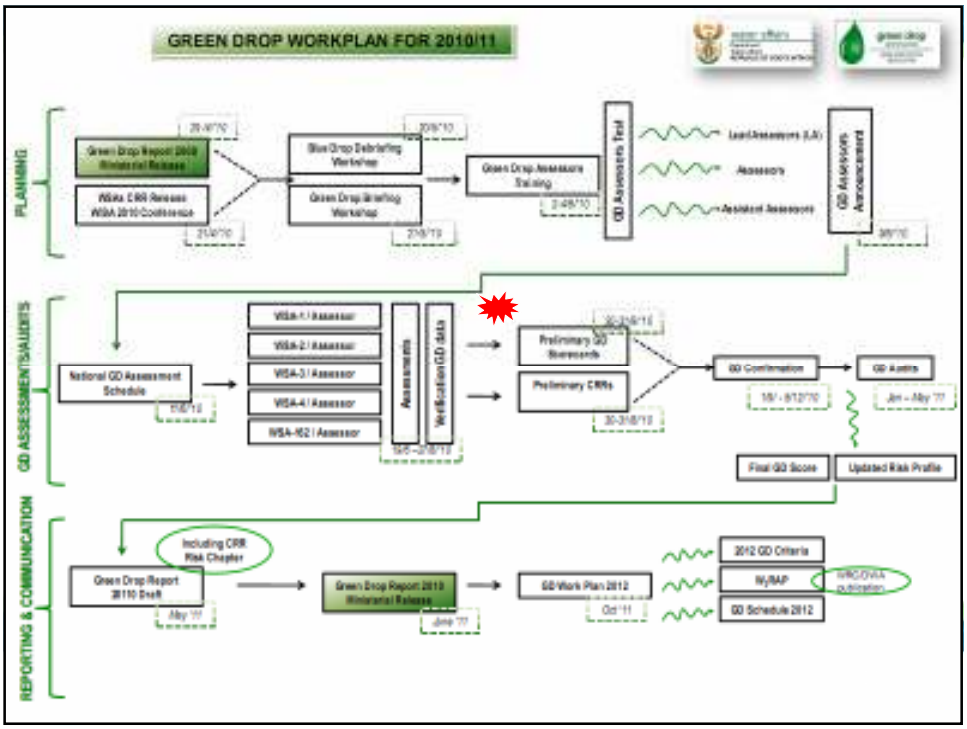
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Green Drop 2011 Assessors Training

- The Department is positioning itself as a credible regulator that will deliver according to international benchmarks.
- 86 DWA officials + 5 Consultants **(101)** were trained as Green Drop Assessors in June:
  - 97 wrote exams
  - 56 qualified to be assessors



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Thank You



We will continue to regain the confidence of the Public in the South African Water Quality and Services fraternity